

Erdős Problem #415

EP-415	L1 - Tractable	Partiellement résolu	Number Theory
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Énoncé original (anglais, reproduction mot pour mot)

For any n let $F(n)$ be the largest k such that any of the $k!$ possible ordering patterns appears in some sequence of $\phi(m+1), \dots, \phi(m+k)$ with $m+k \leq n$. Is it true that $F(n) = (c+o(1)) \log \log \log n$ for some constant c ? Is the first pattern which fails to appear always $\phi(m+1) > \phi(m+2) > \dots > \phi(m+k)$? Is it true that 'natural' ordering which mimics what happens to $\phi(1), \dots, \phi(k)$ is the most likely to appear?

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